

Ibrahim Shoukry

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EDUCATION

University of California, Berkeley

Berkeley, CA

BS, Electrical Engineering & Computer Sciences

Graduating December 2025

Coursework Algorithms, Advanced Programming with C and C++, Internet Architecture & Protocols, Database Systems (DBMS Architecture, Distributed Systems), Computer Security (Cryptography, Secure Systems, Network Security)

TECHNICAL SKILLS

Languages: C++, Python, Go, Java, Javascript, HTML, CSS, SQL

Libraries: Numpy, Pandas, Selenium, NLTK, JSON

Dev Ops: Git, Figma, Google Cloud Platform, Vercel, Jupyter Notebook, ROS (Robot Operating System)

Other: Cloud Computing, Computer Vision, Geospatial Data Analysis, MongoDB

EXPERIENCE

Computer Vision Engineer

May 2024 – Aug 2024

Evitado Technologies

Hamburg, Germany

- Developed a **LiDAR-based system** to detect and segment moving objects near commercial aircraft, enhancing situational awareness for on-ground risk assessment.
- Researched and evaluated a range of **deep learning models** to determine the optimal solution for real-time object segmentation and motion prediction.
- Enhanced an open-source LiDAR processing pipeline by integrating support for custom geospatial data formats (including **rosbags**), ensuring seamless compatibility with proprietary datasets.
- Collaborated with cross-functional teams by presenting technical findings and proposing process improvements, which contributed to refined deployment strategies.

PROJECTS

Counseling Chatbot | Jupyter Notebook, JSON, Numpy, Pandas

[Project Page](#)

- Contributed as part of a diverse team of developers and project managers to create an automated digital counselor that addresses common academic counseling queries for incoming DVC students.
- Engineered a **Machine Learning-Powered Chatbot**: Developed a convolutional neural network using **TensorFlow Keras**, incorporating natural language processing techniques to enable the chatbot to detect, understand, and respond to user questions.
- Integrated JSON for Dynamic Training Data: Utilized JSON data to effectively train the chatbot in recognizing and categorizing questions, ensuring accurate and contextually relevant responses.
- Enhanced New Student Support: The chatbot streamlined access to critical information—such as class choices, schedules, and academic counseling advice—helping students who couldn't meet with a counselor before the semester began, thereby improving initial campus integration and reducing onboarding delays.

UCBMealz | React.js, Selenium, Gemini API, Git

[Devpost](#)

- Engineered a Personalized Meal Planning Solution: Developed a web application that generates custom meal plans for UC Berkeley students by scraping real-time dining hall menus and nutritional data using **Selenium**.
- Leveraged Gemini AI for Advanced Customization: Integrated the **Gemini AI API** to tailor meal recommendations according to individual calorie and macronutrient goals, delivering precise and relevant dietary suggestions.
- Built a Responsive User Interface: Crafted an intuitive, **React.js**-based front end that enables seamless user interaction and simplifies meal planning for busy students.
- Overcame Technical Challenges: Tackled issues related to dynamic and hidden web elements during scraping and managed complex API integrations—all within a tight hackathon timeline—ensuring a fully functional, reliable application.

End-to-End File Sharing System | Go

- Designed and implemented a secure file-sharing server using **Go**, employing industry-standard cryptographic techniques—such as **RSA** for public key encryption, **Argon2** for secure password hashing and key derivation, and **HMAC** for message authentication—to ensure end-to-end encryption, data confidentiality, and integrity.
- Developed a **client API** to enable secure file creation, sharing, access revocation, and modification, ensuring that files remained tamper-proof and resistant to unauthorized access.
- Engineered the system to protect against potential attack vectors, including revoked user adversaries and data tampering, while allowing for efficient file sharing and modification.